

Youssef EL OTMANI

AI & Machine Learning student (M2) · Université Paris-Saclay

Looking for a 6-month internship in AI & machine learning, starting March 2027

@ youssefelotmanipro@gmail.com ☎ +33 7 50 71 57 49 ⌂ Paris, France
in linkedin.com/in/yeot </> github.com/youyou354

youyou354.github.io
Interactive portfolio, with projects you can run

EDUCATION

MSc in Artificial Intelligence 2025 - 2027
Université Paris-Saclay, Faculty of Sciences: taught entirely in English
Year 2 in progress. Year 1: machine learning, deep learning, optimisation, NLP, information retrieval, logic & constraints, a research project and an internship.

Double BSc: Mathematics and Computer Science 2022 - 2025
Institut Galilée, Université Sorbonne Paris Nord, Villetaneuse, France
Mention Très Bien in year one, Bien in years two and three. Class representative, robotics club.

Erasmus exchange semester January - June 2025
Mälardalen University, Västerås, Sweden
Courses in artificial intelligence, computer algebra and computer graphics.

EXPERIENCE

Intern, AI team May - August 2026
Fédération Agirc-Arrco
Four months helping build the federation's AI platform and the tools for specific use cases, and supporting the people who use them.

Private tutor Academic year 2024/2025
Maths, physics-chemistry, biology and computer science
Private lessons for school pupils.

SKILLS

Programming languages
Python · C · Java · OCaml · JavaScript · HTML · CSS · XML · SageMath · OPL · Rocq

Machine learning
PyTorch · Hugging Face Transformers · Keras · scikit-learn · XGBoost · NumPy · pandas · NLP · Deep learning · SMOTE · Codabench

Operating systems
Linux · Windows

Tools
Git · Docker · Azure · Langfuse · OpenTelemetry · Word · Excel · PowerPoint

LANGUAGES

French Native · DALF C1
Arabic Native
English IELTS C1
Swedish A2
Korean A1

CERTIFICATION

Google IT Support Professional Certificate

INTERESTS

Archery · Swimming · Volleyball · Robotics

SELECTED PROJECTS

Personalised learning with a Hierarchical Reasoning Model
M1 research project · team of 4
A video-recommendation platform with two engines, an HRM and a prompted LLM; re-checking our HRM afterwards showed it had learned the wrong target, so it never beat the "always correct" baseline.
Python · PyTorch · HRM · LLM prompting · Gradio

Pollinator identification: a Codabench challenge
M1 · team of 6 · 19/20
A Codabench competition built from 528 INRAE videos: macro-F1 scoring, unseen camera angles in phase 2. I wrote the ingestion and scoring programs and the starting kit.
Codabench · Python · scikit-learn · EfficientNet features

Which small model detects AI inpainting? A benchmark
M2 · team of 5
Ran the training-free detector AEROBLADE zero-shot on 4,459 TGIF images (Stable Diffusion 2, Photoshop), thresholds calibrated by cross-fitting: ROC-AUC 0.505, a clear negative result.
Python · PyTorch · TruFor · PSCC-Net · ForMa

Named entities in French medical text
M2 · pair
17 taggers on the QUAERO corpora: LSTMs and CNNs on six word embeddings against fine-tuned transformers. CamemBERT-bio comes first, F1 68.4.
Python · PyTorch · Hugging Face Transformers · gensim · CamemBERT

Email thread summarization with event triggers
M1 · team of 4
Fine-tuned DistilBART on Enron threads with silver summaries distilled from GPT-4.1 nano and Gemini, with and without event triggers; ROUGE and BERTScore.
Python · Hugging Face Transformers · DistilBART · ROUGE · BERTScore

Does attention help DRAW draw?
M1 · team of 3
An ablation of DRAW's spatial attention on MNIST, EMNIST and Omniglot: an attention grid too small for the data blurs thin strokes; enlarging it closes the gap.
Python · Deep learning · LSTM · VAE

University timetabling with constraints
M1 · pair
A university timetable in OPL / CP Optimizer, from feasibility to three weighted criteria (gaps, heavy days, days on campus), checked on 45 test instances.
OPL · IBM CP Optimizer · Constraint programming · LaTeX

Artificial intelligence labs: Mälardalen
Erasmus
Search (BFS, DFS, A*), a genetic algorithm and ant colonies for the travelling salesman, minimax for Mancala, and an MNIST network at 93.4 %.
Python · Keras